Noroff University College

Bachelor in
Cyber Security

PASSPOWER

Supporting Inexperienced Users (13+) in Creating Secure Passwords and Enhancing Recall through Visual Reinforcement

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Abstract

PassPOWER is a web-based tool developed to help users create stronger, more memorable passphrases. Aimed at people aged 13 and up who may be new to secure password practices, the application generates passphrases, checks their strength, and then transforms them into simple stories and visuals designed to support memory.

The system uses generative AI to turn each phrase into a short scene description and image, offering users a visual reference to help them remember their passphrase more easily. This approach builds on research into visual learning and dual coding, which suggests that combining words with images can improve long-term memory recall.

PassPower demonstrates how AI tools can support password creation in a way that is both accessible and engaging, particularly for younger users who benefit from more interactive and visual learning methods.

Keywords: *Password Strength, Passphrase, Security, Authentication, Youth Digital Safety, Cybersecurity*

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CHAPTER 1

Introduction

1.1 Introduction

As online activity becomes a bigger part of everyday life, the need for secure authentication methods has become increasingly important. From gaming to cloud-based school platforms, inexperienced users are online earlier and more frequently than ever before (Moreno et al., 2013). Despite the rise of alternative authentication methods, passwords remain one of the most widely used mechanisms for protection online (Cain et al., 2018; Lee et al., 2022). However, many less-experienced users continue to misunderstand what makes a password secure, and the importance of online security (Theofanos et al., 2021).

The National Institute of Standards and Technology (NIST), an internationally respected cybersecurity standards body, offers periodically updated guidelines for best practices in password creation (Temoshok et al., 2024, Section 3.1.1.2). In its 2024 publication, NIST raised the recommended minimum password length from 8 to 15 characters and removed the long-standing emphasis on including special characters. Instead, it identifies length and unpredictability (entropy) as among the most important characteristics of secure passwords. While these changes aim to improve usability and security, they also require the memorisation of longer and more complex passwords. Although secure, a password of this length and complexity is difficult to memorise, as humans have a limited ability to recall complex and arbitrary secrets. This limitation is even more pronounced in younger users (Prior & Renaud, 2020; Theofanos et al., 2021).

Recent advances in artificial intelligence (AI), particularly in prompt-based image generation, offer innovative methods for information processing and memory retention. These tools can generate personalised visuals from text input, enabling the potential for image-based, visually cued memorisation of complex passwords. By forming strong associations

with images, users may find it easier to recall secure credentials.

This project explores how educational tools using artificial intelligence (AI) and visual reinforcement techniques can improve password generation and memorisation practices among inexperienced users, particularly those aged 13 and above.

1.2 Problem Statement

As more people, particularly young and less experienced users, become active in an online-dependent world, poor password practices pose critical security risks (Jiow et al., 2021; Theofanos et al., 2021). Many users rely on weak and easily guessed passwords due to a lack of awareness or understanding of secure password creation strategies (Lee et al., 2022; Nisenoff et al., 2023). These poor practices can expose users to exploitation by bad actors, risking threats such as social engineering, dictionary attacks, and brute-force guessing attacks (Guembe et al., 2022).

While generating a lengthy and complex password is simple; memorising a 15-character string of random characters or abstract elements is not (Clark et al., 2024). Traditional password creation methods often emphasise complexity over usability, leading users to follow bad practices like insecure storage of passwords, reusing passwords across platforms, or resorting to insecure simplifications in password creation (Kim & Lee, 2024; Nisenoff et al., 2023).

Additionally, the most widely available Password Strength Meters (PSMs) offer limited educational value (Bitwarden Inc., 2024). They provide static or outdated feedback and assume a level of password literacy that inexperienced users may not yet possess. This lack of support often results in missed opportunities to teach foundational password principles.

Together, these challenges highlight a gap in current password generation and evaluation tools: *they do not adequately support complex password memorability, nor are they designed with the developmental needs of younger or less experienced users in mind.*

1.3 Aims & Objectives

Aim:

The aim of this project is to research, design and develop an educational tool that supports the transfer of secure password practices to password generation and facilitates the memorisation of complex passwords through generative image technology as cognitive cues. The tool will target inexperienced users aged 13 and above.

Objectives:

To achieve this aim, the project sets out the following objectives:

1. Identify common misconceptions about password security in young (13+) and inexperienced users.
2. Develop a secure password generation and evaluation module that meets current industry complexity recommendations.
3. Integrate AI image generation and associated storytelling prompts, to deliver visual and cognitive reinforcement for password memorisation.
4. Evaluate the effectiveness of AI-generated scenes and images in supporting memorisation of secure passphrases for age-appropriate users.
5. Identify and address potential ethical and security risks associated with password and image-generative AI tools.

   Note: The objectives are listed in the order they are intended to be addressed during the development process, rather than in order of importance. Although Objective 4 appears later in the list, it is designed to directly address the research question and represents the primary focus of this research paper.

1.4 Research Question

This study is guided by the following key research question:

> *Can generated passphrases that meet modern security recommendations be securely and effectively represented by age-appropriate, AI-generated images to support memorisation?*

1.5 Scope and Limits

Scope

This project will focus on the design, development, and evaluation of an AI-assisted password generation and memorisation tool for inexperienced users aged 13 and up. The tool is intended to educate users on password security principles through engaging, age-appropriate methods such as gamification, storytelling, and visual reinforcement. It will feature a child-friendly interface and a PSM aligned with the latest guidelines presented by the National Institute of Standards and Technology (NIST). Rather than developing new AI models from scratch, the study use existing models and frameworks, focusing on how they can be applied effectively in a youth-centric education context.

Limits

Although this project aims to create an engaging AI-assisted educational tool for young users, there are certain constraints to consider:

- The tool's ability to generate and evaluate passwords will be limited to established structural guidelines, known weak patterns, and previously exposed credentials. It

will not perform advanced threat modelling or dynamic security analysis beyond these predefined criteria.

- The project focuses on introductory password security concepts. It does not cover more advanced topics such as Multi-Factor Authentication (MFA), biometric authentication, or hardware token-based access.
- Due to time and resource constraints, the gamification elements will not be fully explored. Instead, focus will remain on image-based reinforcement and interactive prompts rather than fully developed game mechanics.
- Live surveys, direct data collection, or public tool testing will not be conducted due to ethical restrictions. Instead, the project will focus on designing the testing procedures and survey instruments, setting the foundation for future research, while relying on peer-reviewed literature and published data sets to inform appropriate design decisions.
- The system will not store or transmit personal data or passwords beyond the user's local environment. All processing will occur in real-time on the user's device to limit the risk of data leakage.
- The project is developed as a proof-of-concept (PoC) and will not be deployed in a real-world environment during this research phase.

1.6 Document Structure

This document is organised as follows: Section 1 presents *Introduction*, establishing the context and motivation for the research. Section 2 provides a *Literature Review*, outlining existing work related to password security, memorisation techniques, and educational tools. Section 3 *Design* details the methodology, including the design and development of the proposed tool. Section 4 *Results*, presents the discussion of preliminary testing results. Finally, Section 5 offers the *Conclusion*, summarizing key findings and identifying directions for future work.

CHAPTER 2

Literature Review

2.1 Background

While technical solutions like rule-based software-generated passwords have simplified the process of creating secure password alternatives, they often overlook one of the root usability challenges: *memorability*. Passwords that are highly secure from technical and security standpoints can be rendered ineffective if the user cannot reliably recall them when needed.

This literature review follows a layered approach to examining this issue. It begins by identifying the characteristics of weak passwords and the behavioural patterns that lead users to adopt insecure practices. It then explores how early cybersecurity education may address these issues, and how secure password generation practices align (or fail to align) with widely accepted NIST standards. Building on this foundation, the review investigates the role of memorability in password usability, with a focus on visual cue-based strategies. Finally, the potential of AI-generated imagery to support password retention is considered, including an assessment of the challenges associated with incorporating AI into cybersecurity tools.

2.2 Inexperienced Users and Weak Passwords

Despite the growing adoption of MFA, password validators, and system-enforced restrictions, many users rely on weak and predictable passwords (Lee et al., 2022; Nisenoff et al., 2023). Common examples include pet names, favourite colours, or *keyboard run* sequences like '123456' or 'qwerty' (Bopp et al., 2024; Cain et al., 2018). These choices may be attributed to limited awareness of security risks and/or preferences prioritising convenience over complexity (Jiow et al., 2021; Theofanos et al., 2021). This tendency

toward simple and easily remembered passwords is evident among young and inexperienced users (Prior & Renaud, 2020). In the case of children, password creation and management are often delegated to parents or teachers, who may themselves default to insecure options out of convenience or a lack of cybersecurity knowledge (Theofanos et al., 2021). As these younger users grow older and engage more independently online, this early exposure to weak password habits can set a problematic precedent.

Research indicates that children often understand, at a basic level, that passwords serve a protective function, yet their actual behaviour does not reflect secure practices (Prior & Renaud, 2020). For example, many children reuse passwords, share them with friends, or construct passwords using easily guessable personal information (Theofanos et al., 2021). They can also underestimate the risks of poor password hygiene due to *optimism bias*, which describes a user's feeling that negative outcomes are unlikely to happen to them personally (Paudel & Al-Ameen, 2024).

Similarly, adults with limited cybersecurity knowledge also tend to display poor password hygiene, such as using simple or repeated passwords across multiple platforms, and in many cases, do so even when they perceive themselves as competent users (Cain et al., 2018). Affected users struggle with maintaining secure password practices due to the overwhelming number of credentials they are expected to manage, a behaviour observed as *security fatigue* (Cain et al., 2018; Furnell & Thomson, 2009; Theofanos et al., 2021). These patterns imply that weak password practices are primarily the result of knowledge gaps and insufficient training, rather than a lack of motivation or concern.

The absence of consistent and appropriate cybersecurity education contributes significantly to the persistence of weak password habits among inexperienced users. Existing educational efforts are often fragmented and not clearly tailored to age-appropriate stages (Sağlam et al., 2023). Without structured and early exposure to essential security concepts, many children develop poor habits based on convenience and trial-and-error rather than informed decision-making. Children often view passwords as obstacles rather than tools for safety, a misunderstanding that stems from insufficient guidance and context (Theofanos et al., 2021). These findings point to the value of educational approaches that are engaging and suited to children's developmental needs.

2.3 Role of Early Cybersecurity Education

Traditional classroom education may not be the most efficient way to teach cybersecurity to children. Alternatively, gamification and storytelling have shown promise in improving understanding and motivation in cybersecurity learning (Bopp et al., 2024; Sağlam et al., 2023). Emerging strategies grounded in interactive learning and family participation have also shown promise. For example, the game-based initiative *CyberFamily*, focused on parent-child collaboration and led to measurable improvements in children's password security knowledge and safe online behaviour (Quayyum & Jaccheri, 2025). Similarly, educational storytelling that embeds password guidance within emotionally meaningful

scenarios, such as the frustration of losing access to a favourite game, or the embarrassment of leaked personal content, has been shown to enhance the internalisation of security principles among young learners (Paudel & Al-Ameen, 2024).

These findings point to the value of playful learning experiences that can embed security practices in younger (and similarly older) audiences. Educational experiences situated within familiar contexts can foster a stronger connection between abstract cybersecurity principles and real-world consequences.

However, for young users, especially those with limited cognitive maturity, introducing complex password requirements poses a significant risk of cognitive overload. If the demands are too high, it can undermine both usability and security, potentially leading users to abandon services altogether due to *registration frustration* (Paudel & Al-Ameen, 2024). For this reason, it is important to support password habits in ways that remain clear and engaging for this age group.

Maintaining this balance requires adaptable educational approaches that can respond to the diverse needs and cognitive capabilities of young users. Advances in AI, including Large Language Models (LLMs), present new opportunities in adapting content for a more interactive and personalised learning experience (Bopp et al., 2024). These systems can deliver personalised instruction, respond to learner questions in real time, and dynamically adjust complexity based on the user's age or skill level. AI-driven applications can also adapt content across different formats, combining textual with generative visual content to improve user engagement.

While these educational approaches play a critical role in shaping secure behaviours early on, they must be supported by technical standards and tools that enforce best practices. Guidelines such as those provided by the National Institute of Standards and Technology (NIST) define essential criteria for password strength, including minimum length, adequate entropy (a measure of unpredictability), and the avoidance of including any Personally Identifiable Information (PII).

2.4 Current Landscapes

2.4.1 Secure Password Strength Guidelines

NIST Special Publication 800-63B, provides widely recognised and detailed recommendations for digital identity management, including password creation practices. The 2024 revision departs from traditional complexity requirements, such as the mandatory use of upper and lower case letters or special characters, instead promoting the use of long user-friendly passwords that are easier to remember but difficult to guess (Temoshok et al., 2024). This approach often involves combining multiple unrelated words into a single password, promoting memorability through meaningful or familiar language rather than abstract character strings. While sometimes referred to as *passphrase*-style passwords, the emphasis remains on creating secure credentials that balance usability and unpre-

dictability.

This approach is especially relevant when considering foundational cybersecurity education. For learners in the early stages of development, it should be emphasised that password strategies support memorability without sacrificing strength. A practical application of this principle is the creation of age-appropriate passwords that also follow NIST guidelines. The resulting passwords should combine length, unpredictability, and memorability through meaningful words or patterns (Temoshok et al., 2024; Theofanos et al., 2021). In this context, *meaning* does not need to originate from the user's prior experiences; it can also be externally introduced. For example, pairing a password with a corresponding image may help users assign meaning to otherwise unrelated words, enhancing memorability through visual association (Paivio, 1979, Chapter 8).

2.4.2 Password Strength Meters

While the NIST guidelines emphasise length, entropy and usability, it is important to consider whether current, popular PSMs and validators assess passwords according to these updated principles. Many platforms continue to promote outdated complexity rules, potentially misleading users about what constitutes a secure password. To explore this further, the following comparison evaluates two password formats using Bitwarden's publicly available strength estimation tool.

Consider the passphrase-style *magic-giraffe-dance*, which uses three unrelated, randomly selected dictionary-based words, and includes hyphens to improve readability. This format adheres to NIST's recommended minimum of 15 characters. When tested using Bitwarden's tool, this password receives an estimated time-to-crack of several centuries (Figure 2.1).

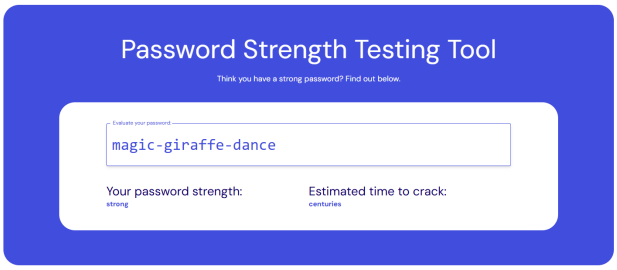


Figure 2.1: Bitwarden password strength tool with stronger password results

In contrast, the password *Giraffe123!*, satisfies several traditional complexity requirements: it contains an upper and lowercase letter, a number, and a special character, and exceeds the commonly enforced eight-character length minimum. However, it includes predictable patterns such as the numeric sequence '123' and terminating exclamation mark, both of which are frequently used to meet minimal complexity criteria with little actual gain in entropy (Ur et al., 2017). When tested using Bitwarden's password strength tool, 'Giraffe123!' receives an estimated time-to-crack of just **four hours** (Figure 2.2), highlighting the disconnect between perceived and actual password strength.

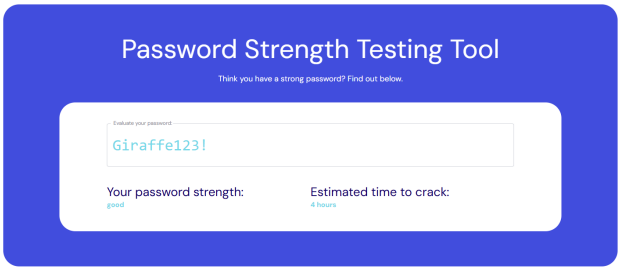


Figure 2.2: Bitwarden password strength tool with weak password results

Despite these weaknesses, this type of password is still widely accepted by major platforms. As shown in Figure 2.3, the top 6 most popular apps among children were identified, and their basic password requirements are presented in Table 2.1. When arbitrary rules, outdated policies and inaccurate guidelines cannot be standardised or trusted, the responsibility for creating secure passwords ultimately falls to the user. Many outdated password policies continue to promote practices that may appear secure but are ultimately ineffective. PSMs and verifiers often rely on outdated rule-based checks, resulting in technically obsolete feedback and misleading strength ratings.

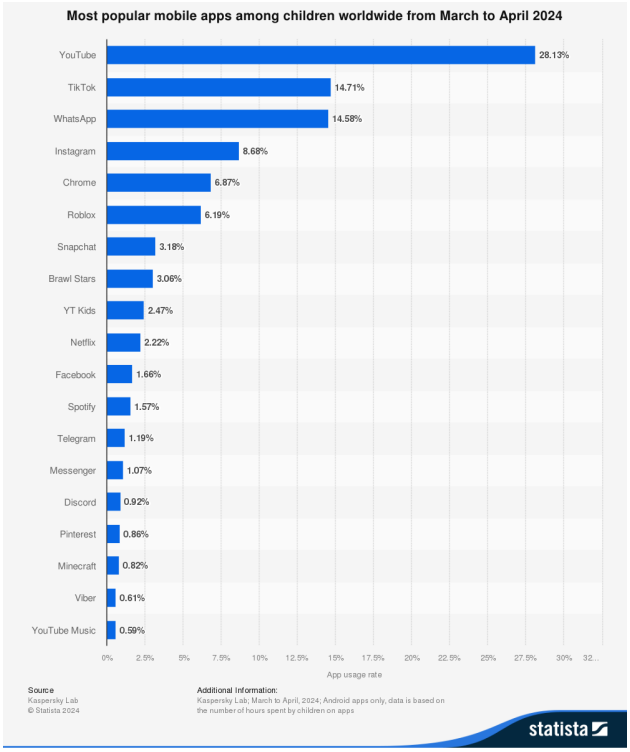


Figure 2.3: Most popular mobile apps among children worldwide from March to April 2024 (Statista, 2024)

Website	Min Len.	Upper/Lowercase Req.	Special Chars Req?	PSM Inc?
YouTube(Google)	8	Yes	Yes	No
TikTok	8	Yes	Yes	No
WhatsApp	6	No	No	No
Instagram	6	No	No	No
Chrome(Google)	8	Yes	Yes	No
Roblox	8	No	No	Yes

Table 2.1: Password requirements collected from official service documentation and sign-up interfaces (Google, 2025 TikTok, 2025 WhatsApp, 2025 Instagram, 2025 Roblox, 2025).

Case Example: Roblox Password Verifier

Roblox, a popular online gaming system aimed at younger users, offers a basic password creation verifier when creating a new user account (Figure 2.4). In this case, the only initially displayed requirement is an 8-character password. Although 'password123' is rejected for lack of complexity, the previously proven weak credential of *Giraffe123!* is accepted, indicated with a green border around the input field.

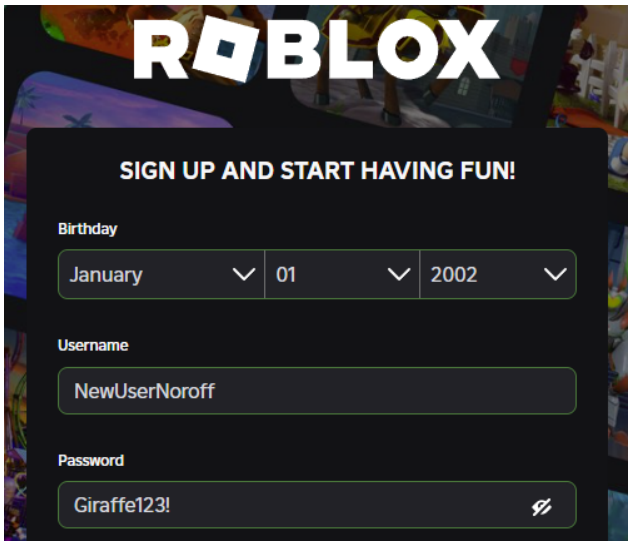


Figure 2.4: Roblox Registration of New User, and Corresponding PSM (2025)

Differing site and software password policies can be confusing for educators and parents. A study of German websites found that some risks were not even mentioned by government-endorsed sites, indicating the need for greater coordination and evidence-based guidance (Bopp et al., 2024).

2.4.3 Big Box Platform Password Validators

Most websites, even major platforms (Lee et al., 2022), still offer only basic PSMs or password strength validators. The outdated policy requirements of mixing punctuation, numbers, and capital letters do little to educate users about the rationale behind the associated platform's password policy's rules and requirements (Umejiaku et al., 2023).

These systems also often fail to evolve with modern attack methodologies, rendering them less effective in mitigating real-world threats such as generative adversarial network (GAN) based attacks, or increasingly sophisticated dictionary-based and pattern-learning attacks (Guembe et al., 2022; Rando et al., 2023).

Validator Algorithms

Bitwarden's password strength tool, in comparison, is powered by *zxcvbn* (Wheeler, 2016), an open-source password strength estimator. Unlike traditional entropy calculators focusing purely on character variety and length, *zxcvbn* incorporates real-world password data, dictionaries, and vulnerable pattern recognition. It can identify common character substitutions, predictable keyboard runs, dates, and known weak structures.

Notably in the scenario of Figure 2.1, *zxcvbn* does not estimate significantly higher entropy for the second password based solely on the inclusion of hyphens. This is because it treats common separators as predictable formatting rather than sources of randomness. As a result, the perceived strength of a password like "magic-giraffe-dance" comes primarily from the entropy and word combination itself, rather than from the inclusion of special characters.

While tools like Bitwarden and evaluators such as *zxcvbn* reflect modern standards for password assessment, the actual implementation of password policies across platforms remains inconsistent, with many popular platforms not meeting NIST-recommended security levels (Figure 2.3).

This inconsistency across platform implementations reveals a critical issue: users cannot rely solely on site-provided validators to gauge password strength accurately or meet recommended security standards. As such, there is a growing need for more dynamic, user-friendly solutions that both generate strong passwords and educate users about why they are secure.

2.5 Effectiveness of Passphrase Generation and Education Tools

While existing password generators can produce strong credentials, these suggestions are difficult to memorise due to their high complexity. For example, Bitwarden offers the option of generating a password or passphrase. As shown in Figure 2.5, the generator can produce a NIST guideline-approved password such as `RP4^JK@Mds9r8oE`. While this string is secure, its randomness and lack of inherent meaning make it challenging for users to memorise.

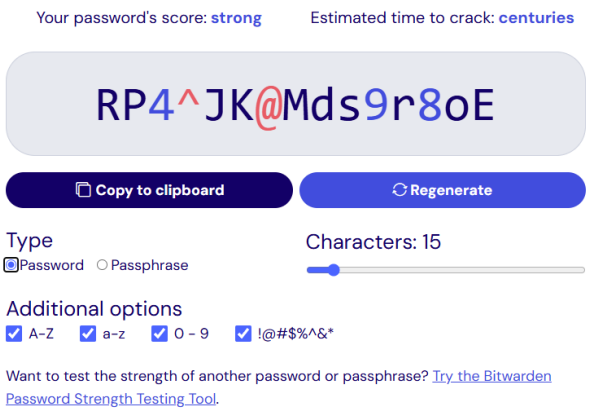


Figure 2.5: Auto-Generated 15 Character password from Bitwarden (April 2025)

Comparatively, the passphrase alternative, Figure 2.6, generates *footsore-doormat-flaccid-useable*. While this format is more memorable and meets security standards, questions arise about vocabulary suitability for users of different ages. Additionally, the generation process remains largely transactional, with users simply clicking a button and receiving a password, with little context or educational value provided.

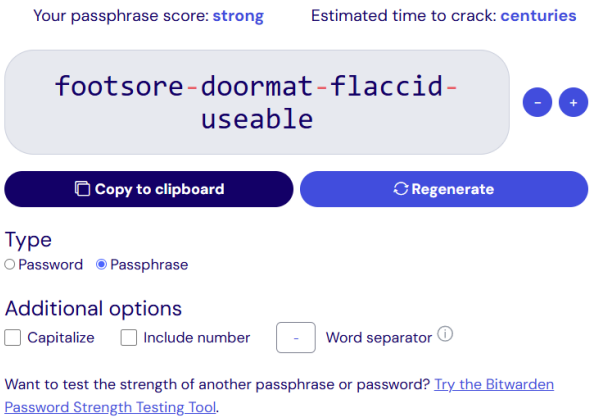


Figure 2.6: Auto-Generated 4 word passphrase from Bitwarden (April 2025)

Integrating an LLM offers promising solutions in this space, assisting with secure password creation and enhancing the education experience around online safety (Paudel & Al-Ameen, 2024; Rack et al., 2024). In theory, AI-assisted tools can provide personalised feedback and real-time guidance, addressing common misconceptions and helping establish more secure password habits from an early age. Studies show that such systems are generally successful in improving the baseline security of passwords, as users generate passwords with greater entropy and resistance to attack models, including AI-powered guessing strategies (Umejiaku et al., 2023).

Despite these strengths, concerns remain about long-term effectiveness. Research indicates that some users revert to unsafe practices once AI support is removed, suggesting a dependency effect that undermines independent security competence (Paudel & Al-Ameen, 2024).

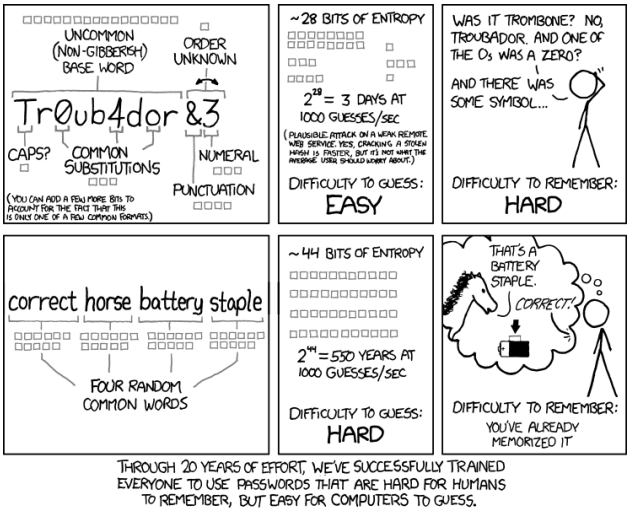


Figure 2.7: “Password Strength” (2011), illustrating how a combination of unrelated words can create a password that is both memorable and highly secure. xkcd / Randall Munroe.

To address this limitation, future design efforts could emphasise cognitive engagement and knowledge retention. Moving beyond traditional generation tools, there is potential in using AI to foster long-term learning through storytelling and interactive education. The next section explores these strategies, examining how they may improve the memorability and meaningfulness of secure password practices.

2.6 Visual Learning and Cue-Based Memorisation Support

One method of improving password memorability is through visualisation. Internally, as a personal imagined scene, or externally as a generated digital image. The use of associative imagery has been shown to assist in recall, as explained by *Dual Coding Theory* (Paivio, 1979, Chapter 7), implying that creative phrases composed of familiar, meaningful components can support memory retention without significantly compromising password strength.

This approach echoes the principle illustrated in the well-known “Password Strength” comic by xkcd (2011) (Figure 2.7). The comic demonstrates that a sequence of unrelated words can offer stronger security and better memorability than a shorter password made up of random characters. Although the comic was originally posted over 10 years ago, the core message, that length and entropy through random words can outperform traditional complexity rules, is still reflected in the most recent NIST guidelines (Temoshok et al., 2024).

This perspective is further supported by Shay et al. (Shay et al., 2012), who suggest that alternative instructional strategies, such as guiding users to visualise their passphrases or construct scenes and narratives using the component words, may prove more effective for improving recall.

These recommendations align with established theories in cognitive psychology. George Miller’s foundational work on working memory demonstrates that individuals can typically

retain around seven discrete items in short-term memory (Miller, 1956). For children and adolescents, this capacity is often lower, which makes remembering complex or arbitrary passwords challenging (Montreuil et al., 2021; Theofanos et al., 2021).

Complementary to this is Allan Paivios's *Dual Coding Theory*, which shows that memory retention is strengthened when information is encoded through both verbal and visual formats (Paivio, 1979, Chapter 7). This suggests that password memorisation techniques may be more effective when they incorporate mental imagery or external visual representations.

Despite these theoretical insights, few current password tools integrate cognitive strategies for visual design features into their user experience.

However, recent developments in generative AI in prompt-based image creation and narrative construction present new opportunities for this integration. AI image generators, such as OpenAI's DALL-E 3, have advanced to the point where they can produce rich visual scenes based on textual descriptions (Betker et al., 2023). In a cybersecurity context, these technologies could be leveraged to generate personalised visual cues that align with user-supplied passwords. Such artifacts may improve memorability while providing an engaging and educational experience, promoting more sustainable password practices over time.

2.7 Effectiveness of Prompt to AI-Image Generation

AI image generation models have become increasingly more advanced, allowing users to create images from simple text prompts. While many studies highlight the visual quality and photorealism, an equally important question is how well the generated images reflect the original prompt (Jamal et al., 2024).

Jamal et al. compared four major AI models in their study: OpenAI's DALL-E, Google's Imagen, xAI's Grok, and Stability AI's Stable Diffusion. The evaluation involved generating images from identical prompts and assessing both image quality and how well each image reflected the original description. They found that models such as DALL-E and Imagen were more likely to produce images that closely aligned with the prompts, even when the outputs were not the most photorealistic. This shows that prompt accuracy is not necessarily tied to visual realism; an image can appear less realistic yet more accurately represent the intended concept (Jamal et al., 2024).

The accuracy of AI-generated images remains one of the most critical and challenging aspects of text-to-image generation. While some models perform well when the prompt includes simple or familiar subjects, the accuracy declines when tasked with representing abstract ideas or complex structures. For instance, in the evaluation of emotional concepts, AI models could clearly express basic emotions like "happiness" or "anger," but struggled with more nuanced emotions like "embarrassment". This highlights that AI models often interpret prompts based on common visual patterns that do not always align with

the user's expectations (Hunkele & Smith, 2024).

Beyond accuracy, AI-generated images are becoming increasingly capable of producing visually striking and memorable content. Advanced tools like the earlier DALL·E 2 model and Midjourney have proven effective at creating strong visual impressions when prompts are sufficiently descriptive (Hussen et al., 2025).

As AI systems demonstrate greater creative abilities, concerns have risen beyond visual quality to include broader ethical implications. While much of the early discussion focused on impacts to creative industries and transparency, increasing attention is now being paid to issues such as the generation of harmful or inappropriate content (Hussen et al., 2025). These evolving ethical considerations are especially important when AI tools are applied to sensitive areas such as cybersecurity education. The use of AI-assisted password generation systems raises significant concerns related to privacy and transparency, which become even more pronounced when the intended users are minors (European Parliament and Council of the European Union, 2016). These concerns and related ethical considerations are further explored in the following section.

2.8 Ethical and Regulatory Challenges in AI-Assisted Cybersecurity Tools for Young Users

Integrating LLM and image generation tools in password creation offers a modern approach to improving password security and enhancing user education and overall experience. However, use of AI-assisted password tools, particularly for minors, raises significant ethical, security and regulatory challenges. These challenges include data privacy concerns, bias in AI-generated content, unintended age-inappropriate content outputs, and potential long-term cognitive effects on users.

Given the established rules regarding children's digital rights, such as the Children's Online Privacy Protection Act (United States Federal Trade Commission, 2013), which restricts the collection of personal information from users under 13 without parental consent, and the General Data Protection Regulation for Kids (European Parliament and Council of the European Union, 2016), which mandates data minimisation and heightened protections for minors, any data processing must be localised and anonymised (Montreuil et al., 2021).

Beyond regulatory compliance, ethical design principles must also guide system development. In particular, AI recommendations should be interpretable, allowing users and their guardians to understand the rationale behind suggested passwords, while methods used to encourage secure behaviour must not compromise autonomy (Theofanos et al., 2021).

Data Privacy, protection of PII

To mitigate privacy risks and uphold data protection standards, many respected secure password tools adopt a client-side architecture where all sensitive operations are exe-

cuted locally within the user's browser. Popular tools such as Bitwarden and the Diceware Passphrase Generator (Reinhold, 1995), generate passwords and phrases entirely client-side, removing the need to transmit or store any sensitive data on external servers. This design approach not only reduces the system's attack surface, but also aligns with the principle of data minimisation as outlined in the General Data Protection Regulation (GDPR) (Article 5, GDPR).

Special protections apply when processing the personal data of minors, who may lack full awareness of privacy risks, consequences, and rights concerning the processing of personal data (GDPR, 2016, Recital 38). The same applies to the involvement of minors in research using AI-enabled password tools, as it requires heightened sensitivity to ethical considerations concerning research data collection (Montreuil et al., 2021).

These heightened ethical standards show the importance of designing cybersecurity education not only to be compliant with privacy regulations but also sensitive to the unique needs of young users. In addition to privacy, other risks arise from the very AI systems used to enhance these tools, including the possibility of biased or inappropriate outputs.

Biases and Inappropriate Content

AI language models and AI image generation tools are trained on large datasets, which may inadvertently reproduce social, political, cultural, or gender-based biases (Moussawi et al., 2024). Such outputs may also include inappropriate language or imagery or reinforce harmful stereotypes, raising concern when these systems are used by or designed for children.

To address these issues, integrating AI models into educational or child-accessible tools must be guided by strict filtering mechanisms and by using curated, diverse, and age-appropriate datasets. For example, sourcing wordlists from children's literature or textbooks could improve vocabulary suitability and cultural relevance. However, due to potential legal constraints around data use, open-access resources such as those provided by The Electronic Frontier Foundation (eff.org) may offer a more practical alternative.

Impact of user interaction with AI generative tools

Another important consideration is the potential for over-reliance on generative AI. As children grow up with constant access to AI-powered assistance, there is a risk of diminished cognitive engagement and reduced development of problem-solving or memory-related skills. This phenomenon, sometimes referred to as *intellectual deskilling*, raises questions about the long-term effects of offloading cognitive tasks to machines (Machidon, 2025). While AI can undoubtedly support learning and improve usability, its design must also encourage active participation and critical thinking, rather than passive consumption.

2.9 Related Works

Bitwarden: Offers both password and passphrase generation options, but offers little contextual feedback or support for memorisation.

Diceware: Randomly selected words to produce high-entropy passphrases. Relies on a static wordlist and does not provide personalisation, visual cues, or guidance for different age groups.

correct-horse-battery-staple comic by xkcd: Influential in shifting public understanding of password design. Unrelated words can form a password that is strong and memorable.

2.10 Summary & Gap Identification

This literature review has highlighted several critical gaps in current password education and generation tools. Despite advancements in password policy, users often struggle to apply secure practices in real-world contexts because the tools they rely on offer limited guidance, unclear feedback, and little support for memorability. Early cybersecurity education has shown promise, especially through gamification and storytelling, but these approaches remain fragmented and are often unsupported by the tools that users interact with regularly.

Modern password generators can produce technically strong credentials but often lack memorability and educational value. Passphrase approaches that follow NIST guidelines offer a better balance of security and usability, yet they are rarely presented in a meaningful context that promotes understanding. Visual strategies, such as those grounded in the *Dual Coding Theory*, offer a valuable but underused opportunity to support memorisation.

Recent advancements in AI, including language and image generation models, introduce new possibilities for improving the creation and recall of secure passwords. At the same time, these technologies raise concerns about data privacy, bias, inappropriate content, and the potential for users to become overly dependent on automated assistance. Most current tools do not adequately address these ethical and developmental considerations.

These findings point to the need for a more integrated and user-centred approach. A system that combines AI-generated passphrases with personalised image-based scenes can help users create passwords that are secure, memorable and educational. The following section presents the design and methodology of *PassPOWER*, a prototype developed to address the gaps identified in this review.

CHAPTER 3

Methodology (Design)

3.1 Design

Building on the findings presented in Chapter 2, this research proposes the design of a tool that integrates a complex passphrase generator, an embedded PSM, and a generative visual memory assistant. While the term *password* is often used broadly, this study focuses specifically on *passphrases*, multi-word authentication strings that offer superior entropy compared to traditional passwords.

In line with privacy and security best practices, the tool was designed to minimise data exposure. The default wordlist-based implementation of the passphrase generator runs entirely on the client side, avoiding the need to store or transmit sensitive user data. Alternatively, a second implementation using OpenAI's GPT-4-Turbo (OpenAI, 2025) API to dynamically generate passphrases. While this method allows for greater control over age-appropriate word choice and potentially more memorable outputs, it introduces a higher privacy risk due to the external API call, even though data is protected via HTTPS.

To support these parallel approaches, two passphrase generation modules were developed. The first uses a securely curated, embedded wordlist accessed through Python's built-in `random` module. Though not cryptographically secure, this approach is sufficient for demonstrating proof of concept and aligns with Diceware-inspired entropy principles outlined in Chapter 2. The second method relies on GPT-4-Turbo module to generate passphrases in pre-arranged form, offering variety and contextually meaningful combinations beyond a set wordlist.

Although the comparison of the two alternative designs was not the focus of the memorability evaluation, it remains a useful reference point for future research.

For the purposes of this paper and the proof-of-concept implementation described, the

AI-assisted **chatGPT-4-Turbo method** was used as the default approach for passphrase generation.

The limitations and challenges discussed previously highlight four primary issues that inform the design approach:

Issue 1: Predictable Patterns and Lack of Entropy

Despite the recommendation of including various character types, such as uppercase and lowercase letters, punctuation marks, and numerical digits, many users will unwittingly use weak passwords. Modern brute-force attacks, using common and leaked password databases, can crack these combinations in a limited time.

Proposed Solution: Investigate AI-assisted passphrase generation as a potential method of dynamically producing randomised passphrases.

Issue 2: Usability and user experience (UX) associated with Password Policies and PSMs

Overly complex or outdated password policies combined with ineffective PSMs negatively impact user experience. Users frequently experience frustration and fatigue when interacting with password creation requirements that feel arbitrary, confusing, or overly restrictive. This can lead to disengagement, insecure workarounds, or reduced compliance with secure authentication practices.

Proposed Solution: Design a user-friendly interface that simplifies secure passphrase creation by providing clear and up-to-date cybersecurity guidelines and practical tips. The tool's minimalistic and intuitive passphrase and image generation process is intended to reduce security fatigue and promote the development of more secure online habits. To improve password evaluation, the embedded PSM employs the *zxcvbn* algorithm to assess real-world pattern strength, supplemented by additional rule-based checks such as a minimum length requirement and prevention of duplicate word selection within generated passphrases.

Issue 3: Limited Early Cybersecurity Education

A lack of foundational, age-appropriate cybersecurity education contributes to poor password habits, leaving users ill-equipped to understand or value secure authentication behaviours.

Proposed Solution: Integrate an educational component into the tools, featuring universal cybersecurity tips and interactive quiz modules. These elements aim to provide users with foundational knowledge about secure password practices in an engaging and accessible format. By offering dynamic feedback, the tool encourages users to internalise best practices and build long-term security habits, addressing the educational gaps identified in Chapter 2.

Issue 4: Balancing Password Complexity with Reasonable User Memorization Capacity

Current guidelines recommend passwords of at least 15 characters, *optionally* including special characters and numbers. However, such requirements often exceed typical human memorisation capabilities.

Proposed Solution: Employ an AI-image-generation module as a memorization aid, creating personalised visual representations of generated passphrases to enhance long-term memory recall. As discussed in Chapter 2, extensive research indicates that visual memory typically outperforms verbal memory, making it a valuable tool for encoding complex information such as passphrases. Associating each passphrase with a personally meaningful image is intended to strengthen memorability and support more reliable recall over time.

3.2 Implementation

To address these issues, the application design is structured into three stages.

Stage 1: Password generation and strength evaluation.

Secure passphrases are generated using an AI-assisted method, followed by immediate strength evaluation using the *zxcvbn* algorithm and custom rule-based checks.

Sufficient Entropy: Protection Against Dictionary Attacks

With a 7778-word list, a five-word randomly generated passphrase yields approximately 64.7 bits of entropy, thus meeting the 64-bit minimum entropy requirement set by NIST for session secrets (SP 800-63B,§7.1) To reach this level of entropy, the following example calculation is provided:

Wordlist: 7776 words (EFF's "long word list")
Passphrase: 5 words

$$\text{Entropy per word} = \log_2(\text{wordlist size})$$

Possible Combinations :

$$7776^5 = 2.8318 \times 10^{19}$$

$$\text{Total Passphrase Bits of Entropy} = \log_2(7776^5) = 5 \cdot \log_2(7776) \approx 64.7 \text{ bits}$$

Although the entropy achievable through a five-word passphrase from a large wordlist satisfies the recommended standards, the initial implementation of this application makes deliberate concessions based on usability and user demographics. Two passphrase generation methods were developed.

1. A shortened, curated, and embedded wordlist designed for younger users prioritises

word familiarity, simplicity, and ease of translation into the visual medium over maximum theoretical entropy. This reduced wordlist inevitably lowers the possible entropy of the generated passphrases, but it aims to foster early secure password habits without overwhelming users with complex vocabulary.

2. An AI-assisted passphrase generation method using ChatGPT-4-Turbo, where prompts are structured to dynamically select diverse and age-appropriate words. While this method potentially draws from a broader and richer vocabulary than a shortened, fixed list, it introduces other challenges, such as reduced control over prompt interpretations and output consistency, and additional security considerations due to the transmission of prompts to an external API.

The ChatGPT-4-Turbo assisted passphrase and scene generation method was selected for the initial user survey and proof-of-concept evaluation. This approach allowed for greater variety and narrative richness in the generated materials, aligning with the study's focus on assessing the effectiveness of visual artifacts for memory reinforcement. Figure 3.1 shows a sample output generated during the testing phase, including two example passphrases, their corresponding scene descriptions, and the associated AI-generated imagery. Table 3.1 shows the generated scene descriptions used to prompt the AI image generation module, a character count, and the `zxcvbn` algorithm's calculated score. DALL-E 3 produced the corresponding visuals.

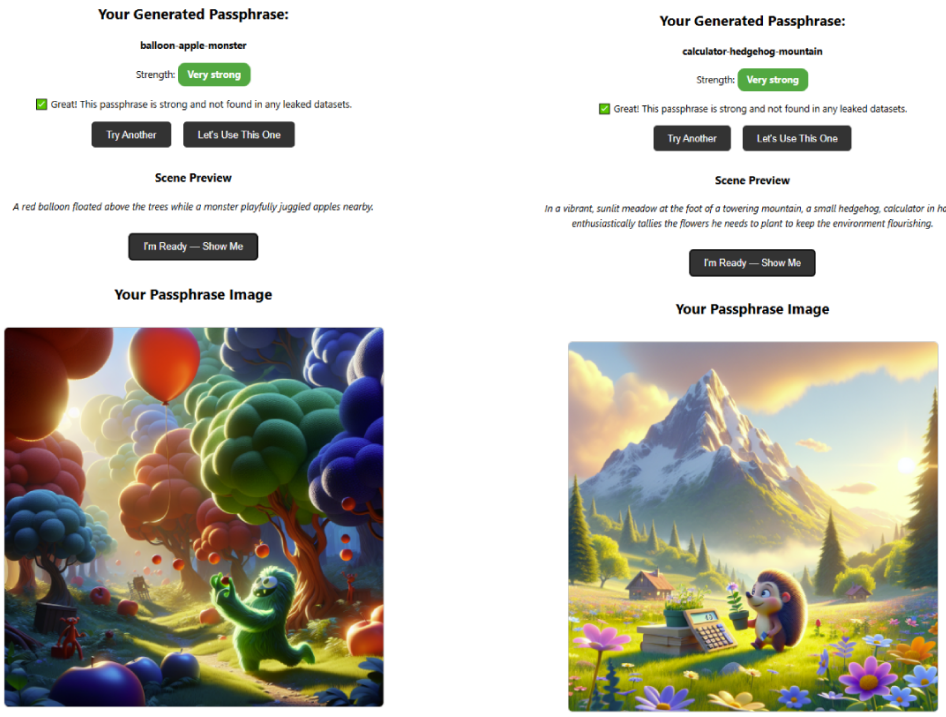


Figure 3.1: Examples of generated passphrases (*balloon-apple-monster* and *calculator-hedgehog-mountain*), corresponding scene descriptions and associated AI-generated imagery.

Generated Passphrase	Generated Scene Description	Char. Count	zxcvbn Score
`balloon-apple-monster`	A red balloon floated above the trees while a monster playfully juggled apples nearby.	19	4/4 Very Strong
`calculator-hedgehog-mountain`	In a vibrant sunlit meadow at the foot of a towering mountain, a small hedgehog, calculator in hand, enthusiastically tallies the flowers he needs to plant to keep the environment flourishing.	26	4/4 Very Strong

Table 3.1: Passphrases, corresponding scene descriptions, character counts, and strength evaluation results using `zxcvbn`. Character counts exclude hyphens.

Sufficient Length: Protection from Brute-Force Attacks

A core design principle of this app is the generation of passphrases with a minimum length of 15 characters. This baseline was selected to ensure sufficient entropy to resist brute-force attacks, assuming modern cryptographic safeguards are in place during storage (OWASP Foundation, 2024). A 15-character passphrase constructed from randomised dictionary words offers a large key space, significantly increasing the computational effort required for exhaustive search attacks.

However, password length alone does not guarantee full protection. Passphrase strength must be considered in tandem with server-side credential handling. The effectiveness of a passphrase is ultimately conditional upon its storage using secure cryptographic techniques, namely the use of salted hashing and Key Derivation Functions (KDFs). Bitwarden, previously introduced in Chapter 2 as an open-source password generation and management solution, can be an illustrative example of secure credential storage. Bitwarden employs Argon2id (Biryukov et al., 2016) for key derivation and implements a zero-knowledge architecture, ensuring that even in the event of a server compromise, the computational cost required to perform offline brute-force attacks remains prohibitive (Bitwarden Inc., 2024).

Stage 2: Password Scene Generation

To avoid sending the passphrase directly in its raw format (e.g., *word1-word2-word3*) to the image generation module, the system constructs a descriptive scene incorporating the core elements of the passphrase (Figure 3.2). This approach serves several purposes:

```
prompt = (
    f"Write one simple sentence describing a scene for ages 13 to 18, "
    f"based on the words: '{words[0]}', '{words[1]}', and '{words[2]}'. "
    f"Use simple vocabulary, Only return the sentence."
)
```

Figure 3.2: Backend prompt used to guide AI scene generation.

- **Obfuscation of Sensitive Data:** Transforming the passphrase into a descriptive

scene reduces the risk of exposing the exact passphrase during AI image generation. According to OpenAI's documentation, input prompts may be temporarily stored on their servers for security and abuse monitoring purposes, making direct transmission of sensitive information inadvisable.

- **Enhanced Image Quality:** Providing a full scene description results in richer, more structured images compared to using raw, isolated passphrase words. This method also reduces the likelihood of the generated image visually representing the passphrase too literally, such as by directly displaying the individual words in an obvious arrangement.
- **Improved Memorability:** Embedding passphrase elements within a narrative structure supports stronger memory encoding by creating meaningful associations.

Stage 3: Scene Description to Image Generation

Following the generation of scene descriptions, these prompts were packaged and relayed to OpenAI's DALL-E 3 image generation model to produce corresponding visual artifacts. To ensure a consistent, child-friendly visual style across all outputs, a standardised backend style prompt was applied to each scene description before submission, as shown in Figure 3.3.

```
style_instruction = (
    "Style: colorful 3D animation, cartoon-like, Pixar or Disney-inspired. "
    "Avoid photo-realism. Use smooth lighting, expressive characters, and child-friendly design."
    "Add extra elements to enrich the scene."
)
```

Figure 3.3: Backend style instruction used to guide consistent AI image generation aesthetics.

While the initial backend prompt structure provided a consistent and effective baseline for image generation, it is recognised that further refinement may be necessary. With expanded survey testing and user feedback, the prompt could be adjusted to better address potential issues. Ongoing refinement of the prompt would aim to ensure consistent output quality, reduce the risk of unpredictable generation artifacts, and support sustained user engagement through appealing visuals.

Stage 4: Educational Component

Following the generation of the passphrase and during the processing of associated imagery, the application transitions into an education module designed to reinforce cybersecurity best practices. This module delivers brief tips regarding secure password habits, passphrase construction, and memorisation strategies. Figure 3.4 shows an example layout of the education and quiz module, including the tips and knowledge reinforcement screen (left) and the quiz interface (right)[1].

[1]The educational component will not be included in the initial testing and survey phase. The survey focused solely on the memorability of generated passphrases, with or without accompanying imagery. Full evaluation of the educational component's effectiveness is reserved for future research.

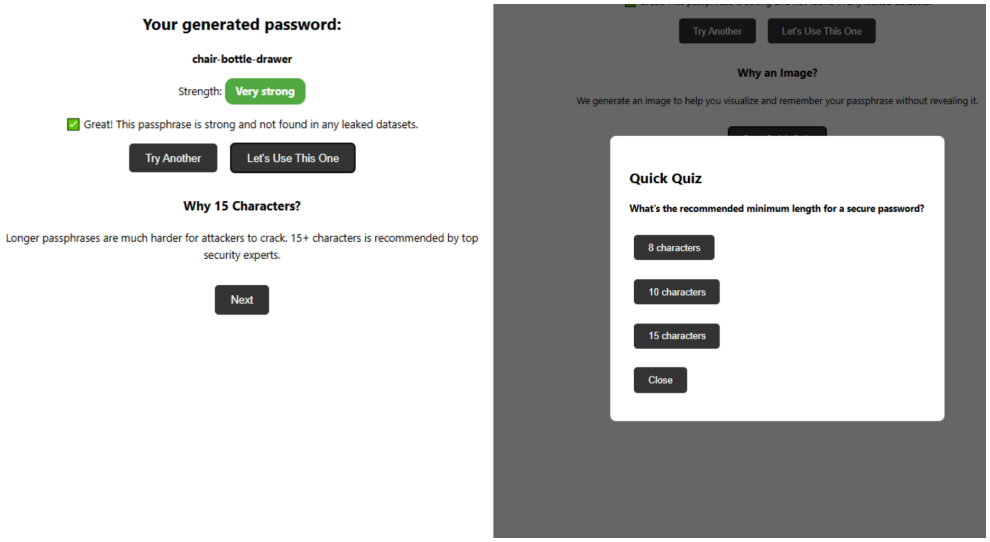


Figure 3.4: Example of educational component: Tips and knowledge reinforcement (left) and optional quiz interaction (right).

3.3 Proposed Survey

This study outlines a proposed survey methodology for future empirical testing, aimed at evaluating the impact of visual and narrative cues on password memorisation. Although no live participant testing was conducted during this phase, the structure provides a clear foundation for later implementation [2].

Participants will be drawn from a volunteer sample aged 18 to 30, serving as a proxy for the tool's intended target group of users aged 13 and above. This adjustment is made due to ethical restrictions on testing with minors.

Each participant will be randomly assigned to one of two groups, receiving a system-generated passphrase designed to meet current security standards:

Group A – Passphrase, Narrative and Visual

Participants in Group A will receive a three-word passphrase, accompanied by a brief, AI-generated narrative sentence that places the passphrase within a contextual scene. Additionally, they will be provided with an AI-generated image visually depicting the scene. Participants will be instructed to memorise the passphrase and the associated content and discard the textual components, retaining only the image to support later recall.

Group B – Passphrase Only

Participants receive the same passphrase but no scene description or image. They are asked to memorise the passphrase, without any supplied narrative or visual reinforcement.

[2] Templates for Consent Form and Survey can be found in Appendix A, B and C

Note: Further experimental groups could be introduced in future testing phases, exploring variations such as narrative-only, image-only, or combinations of different prompt styles and image formats. These comparisons may help isolate which elements contribute most to memorability and user engagement.

Recall Procedure Password recall will occur at three intervals: 24 hours, 72 hours, and one week following initial exposure. Group A will be permitted to view the AI-generated image during each recall session, while Group B receives no external cues at any stage. All participants will enter their recalled passphrases through a standardised interface. Data will be collected on recall accuracy, time taken, and character-level errors. Additionally, Participants will complete short questionnaires to provide subjective feedback on memorability and perceived difficulty.

Although this testing is deferred, technical evaluation confirmed that the system can generate strong, unique passphrases and produce scene descriptions and images that align with each passphrase. This framework ensures the survey can be executed seamlessly in a future phase and provides the basis for assessing how visual and narrative support can impact password retention.

CHAPTER 4

Results

4.0.1 Password Generation Results

As mentioned in the previous Chapter, two primary passphrase generation methods were developed during the initial design implementation:

1. **Local wordlist-based generator**: This method uses an embedded list of age-appropriate words, with passphrases formed by selecting three random entries using Python's `random` module. It operates entirely client-side, offering control, privacy, and simplicity, though its smaller vocabulary limits entropy and diversity.
2. **AI-assisted generator using ChatGPT-4-Turbo**: This approach used structured prompts to generate passphrases via OpenAI's LLM API, ensuring appropriate vocabulary, consistent formatting, and increased variation. While this improved phrasing and uniqueness, it introduced reliance on external infrastructure but reduced transparency in how word choices were made.

The initial goal of the passphrase generation module in PassPOWER was to implement an age-appropriate, client-side solution using a curated wordlist. However, while this approach offered better control and offline security, it presented limitations in vocabulary variety and entropy. After multiple early tests, repetition and reduced word diversity occasionally compromised passphrase strength and memorability.

The alternative, a prompt-trained AI-assisted model, allowed for more dynamic passphrases. Prompt training was necessary for passphrase formatting, appropriate word selection, passphrase length and avoiding potential word repetition. In 100% of the test cases (Figure 4.1), the AI-assisted app met or exceeded 15 character length, used suitable vocabulary for users aged 13+, and provided unique passphrases[1]. These results are reflective of the project's defined scope and objectives (see Chapter 1).

[1] High resolution table found in Appendix D

When the generated passphrases were compared to Bitwarden's example *footsore-doormat-flaccid-useable* (as shown in Figure 2.6), PassPOWER's outputs were found to be generally more age-appropriate and easier to visualise. Bitwarden's generated passphrase included uncommon or abstract vocabulary, potentially affecting simple visualisation and eventual memorability. In comparison, PassPOWER's generated passphrases, for example *calculator-hedgehog-mountain*, used common, image-friendly words that easily translated to scene descriptions and visual association.

This advantage in visualisation ability and vocabulary relevance directly supports the project's objectives, which emphasise memorability in combination with password strength.

#	Generated Passphrase	Character Count (Not including hyphen)	Generated Scene Description	zxcvbn Score (0–4)	Generated Image	Notes
1	backpack-cookie-giraffe	21	In a sunlit savanna, a curious giraffe peeks into a teenagers backpack, gently snoathcing a cookkie with its long tongue, much to the delight of the surrounding giggling children.	4		Clear image. Giraffe and cookie well depicted. Backpack visible but detail could be sharper. Strong scene match.
2	balloon -apple-monster	19	A red balloon floated above the trees while a moster playfully juggled apples nearby.	4		Fun, vibrant scene. Monster and apples clear. Balloon present. Minor artistic abstraction.
3	calculator-hedgehog-mountain	26	In a vibrant meadow at the foot of a towering mountain, a small hedgehog, calculator in hand, enthusiacitally tallies the flowers he needs to plat to keep the environment flourishing.	4		Hedgehog and calculator clearly shown. Mountain present. Vibrant composition.
4	guitar-Kraken-Brocolli	20	At the beach campfire, Jamie stremmed his guiar while everyone joked about a kraken hiding among the broccoli in their dinner salads.	4		Guitar clear. Kraken and broccoli represented, though kraken could be misinterpreted. Kraken noted as a potential unknown object.
5	hamburger-dolphin-notebook	24	At the sunlit pier, Max scribbled in his notebook about the dolphin he'd just fed a hamburger, both of them sharing a magical moment of friendship.	4		Dolphin and hamburger clearly rendered. Notebook was small but visible.
6	helmet-tiger-garden	17	Wearing a helmet, Jake cautiously entered the garden where the circus tiger had been spotted.	4		Helmet and tiger very clear. Garden was not as clear, needs user interpretation.
7	penguin-flower-elephant	21	A penguin waddled past an elephant sniffing a colorful flower in the zoo.	4		Penguin and elephant and flower were all clear. Good clarity and word-to-image representation.
8	Saxophone-Goblin-Triceratops	26	At the funky forest party, a goblin skillfully plaed the saxophone while riding on a friendly triceratops.	4		Wild scene with good clarity. Goblin and triceratops and saxophone distinctly represented.
9	moth-subway-ticket	16	A moth fluttered around the dim light as the teenager held his subway ticket tightly, waiting for the train.	4		Confusion on subway car. Abstract representation, garden growing inside subway car. Ticket and moth represented clearly.
10	sunset-waffle-monkey	18	As the sunset painted the sky orange, a mischievous monkey snatched a waffle right out of a teenager's hand.	4		Monkey and waffle very clear. Sunset background vibrant. All elements present and well rendered.

Figure 4.1: Evaluation of PassPOWER Passphrase-to-Scene/Image Generation (10 Examples). Evaluation conducted April 30, 2025 using OpenAI ChatGPT-4-Turbo for passphrase and scene description, and DALL·E 3 for image generation.

4.0.2 Password Evaluation Results

Each generated passphrase was evaluated for strength and uniqueness using the open-source *zxcvbn* estimator API. In the test (Figure 4.1) of 10 generated phrases, 100% of the samples received the highest rating of 4/4. According to *zxcvbn* documentation, 4/4 is defined as **"very unguessable: strong protection from offline slow-hash scenario"**, meaning the passphrase offers robust resistance to deliberate and prolonged cracking attempts using stolen password databases or hashes (Wheeler, 2016).

In addition, each passphrase was individually checked against the *Have I Been Pwned* (Hunt, 2025) database of known breached credentials. None of the generated phrases appeared in the dataset, confirming that the sample PassPOWER outputs are not present in any large-scale credential leaks.

4.0.3 Scene Description Results

Following passphrase generation, each phrase was transformed into a descriptive scene using the ChatGPT-4-Turbo module. The script embedded the three component words of each passphrase into a short narrative that would support image generation and assist in scene memorability. Prompt engineering was used to encourage controlled outputs, with an emphasis on age-appropriate content and vocabulary (See Chapter 3 Figure 3.2).

In all tested cases, the generated scene descriptions successfully included all three core passphrase elements. However, while word inclusion was consistent, not all descriptions were equally effective in supporting memorability. In several scenes, the language becomes overly elaborate or introduces abstract details that may complicate recall. For example, in Test #1 (Fig 4.1), with passphrase *backpack-cookie-giraffe*, produced the associated scene description below.

> *"In a sunlit savanna, a curious giraffe peeks into a teenager's backpack, gently snatching a cookie with its long tongue, much to the delight of the surrounding giggling children."*

Although all the passphrase elements have been introduced in the scene description, the scene adds additional imagery such as "giggling children", which is unrelated to the core passphrase. Inclusion of the complete narrative, "much to the delight of giggling children," may add unnecessary noise that distracts from the intended memorability value. This tendency toward narrative over-complication was observed in five of the ten test samples, specifically numbers 3,5,6,8 and 10.

Image Generation Results

The final and most significant phase of the research, directly relating to the research question *"Can generated passphrases that meet modern security recommendations be securely and effectively represented by age-appropriate, AI-generated images to support memorisation?,"* The PassPOWER application was evaluated on its ability to visually represent passphrases using generative AI.

The image generation process leveraged OpenAI's DALL-E 3 model to transform the previously generated scene descriptions into clear and engaging visuals. The backend prompt (Figure 3.3) was designed to guide outputs towards child-friendly, animated movie-inspired scenes and away from photo-realism.

Across all test cases, 100% of the resulting images contained clear visual representations of all three passphrase words (see Figure 4.1). No image displayed text or characters that revealed any elements of the original passphrase as a baseline for basic security. As requested, the visuals were age-appropriate for users aged 13 and up and demonstrated a consistent style and tone [2].

Some limitations were observed in how certain generic interpretations were interpreted

[2] A high-resolution set of all generated images is available for reference in Appendix E

and rendered. For example, the word *"Kraken"* in Test 3 was visualised as a two-headed beast resembling a plant creature mixed with a dinosaur, rather than a recognisable sea monster.

Additionally, Test 9 produced the only image in the testing set that displayed an abstract visual style. The input scene description is as follows:

> *"A moth fluttered around the dim light as the teenager held his subway ticket tightly, waiting for the train."*

The resulting image combined the subway platform and train car into a single frame, with greenery and flowers growing from the floor. Although all three passphrase components were represented with *moth-subway-ticket*, the overall layout and environmental blending produced the only example of clear abstractions among the ten image generation tests.

4.1 Summary of Results

The PassPOWER application successfully generated passphrases, scene descriptions and corresponding AI-generated images. Each passphrase met the modern NIST security recommendations for minimum length and entropy, and was validated as strong and unique using the *zxcvbn* estimator, and found no matches against the Have I Been Pwned breach database. Scene descriptions consistently incorporated all core passphrase components, occasionally adding additional narrative details while remaining appropriate for the intended test age group of 13 and above. The resulting images provided visualisations that accurately represented the input scene descriptions, though in some cases demonstrated a capacity to render out-of-prompt-scope abstractions or stylistic interpretations.

Building on these findings, the following section discusses the implications, limitations, and broader relevance of the PassPOWER approach.

4.2 Discussion

4.2.1 Bringing Passphrases to Life

The primary research objective was to investigate whether passphrases that meet modern security recommendations could be effectively represented by age-appropriate, AI-generated images to support memorisation. Inspiration was drawn from the previously mentioned *correct-horse-battery-staple* comic (see Figure 2.7), with this project seeking to bring that idea to life by embedding passphrases in AI-generated imagery. Development and internal testing confirmed the technical feasibility of the PassPOWER approach and its alignment with the project's core objectives, particularly the use of AI-generated visuals to support memorisation for younger or inexperienced users.

Research conducted in the literature review highlighted several persistent factors contributing to poor password creation, particularly amongst these users. These included a tendency to prioritise convenience over complexity, reuse of common words or patterns, and a lack of clear understanding about what makes a password secure. In response, the design decision was made to first implement a secure passphrase generator. This solution was intended to limit the likelihood of users unintentionally including PII, the continued use of weak or familiar passwords, and the development of insecure password habits.

4.2.2 Security and Strength of Generated Passphrases

The passphrase generator was designed to produce strong, randomised passphrases using age-appropriate vocabulary and clear formatting. While a wordlist-based approach was initially exported, the lack of a sufficiently large and audited wordlist limited its practical use. As described in Design (Chapter 3), the AI-assisted method was selected for testing due to its flexibility and ability to generate diverse, appropriate outputs. The results showed that every passphrase generated met our security requirements in length, complexity and uniqueness.

One notable limitation with the AI-assisted model was the unknown size and composition of the underlying "wordlist" used by the ChatGPT-4-Turbo module. Because the effective wordlist is not transparent or user-defined, it is impossible to calculate entropy empirically or ensure consistency across sessions. This lack of control complicates formal strength evaluation and reproducibility. Preferably, a fixed, embedded wordlist of known size would allow for direct entropy calculations. Based on calculations shown in Chapter 3, reaching the recommended 64-bit entropy security level requires at minimum five randomly selected from a list of approximately 7700 unique words. As the current implementation relies on an unknown wordlist and an algorithm-based randomisation, LLMs may not consistently meet this benchmark. This emphasises the need for a larger, embedded, audited wordlist in future versions to support security and measurable compliance with established standards.

Evaluation of generated passphrase strength through *zxcvbn* API provided a practical method in visualizing and interpreting strength score; however, it is not without limitations. The estimator does not calculate true entropy, instead, it relies on pattern-matching, substitution rules, and comparisons against a fixed set of common words and structures. While it remains useful as a comparative baseline, future versions of PassPOWER may benefit from incorporating custom entropy calculations from updated breach data for a more accurate evaluation.

4.2.3 Narrative Scene Generation and Memorability

A key innovation in PassPOWER was the integration of AI-generated storytelling, as scene descriptions, to support memorisation of passphrases and eventually input prompts

for image generation. Drawing from Paivio's conclusions on his Dual Coding Theory, the tool embeds each passphrase into a short narrative intended to give the phrase context, easing visualisation and potentially allowing easier recall of keywords.

During testing, these scene descriptions consistently included all three core passphrase words, with most outputs remaining grammatically consistent and appropriate for the intended age group. However, some scenes occasionally included additional narrative elements or overly descriptive language that could reduce memorability by introducing unnecessary information. Prompt engineering could mitigate this in future versions, but these findings highlight a need to balance creative engagement with clarity and focus, particularly when designing for younger users.

While the use of AI-generated memorisation techniques had not been found in other password tools during the research phase, this approach aligns with broader trends in gamified education and cognitively supportive design. By turning password generation into a creative and story-driven activity, users may develop stronger engagement and memory retention, especially when the tool adapts to the user's level of skills, age and experience.

At this stage, the AI model is first given interpretive freedom. Introducing creative potential and risk. This is the earliest point in the process where the model can interpret commands more broadly, potentially outputting irrelevant, biased, or age-inappropriate content. Although prompt engineering and internal testing helped guide outputs towards more focused narratives, this remains a key area for ongoing monitoring. Any future implementations should include mechanisms for regular prompt audits, content review, and model behaviour tracking to ensure outputs remain safe, relevant and effective for the target age of the user group.

4.2.4 Image Generation as a Visual Memory Aid

The image generation stage, powered by OpenAI's DALLE-3, represented the third and most visible layer of AI intervention. While the majority of outputs (see Appendix D) maintained a clear and consistent age-appropriate style, some images contained unexpected abstraction or visual representation beyond what was described in the scene prompt. Some scenes blended elements, while not inappropriate, could confuse or distract users, taking away from the core purpose of the exercise.

It is also important to clarify that the scene descriptions used to generate the images are not intended to be stored by the user. In deployment, only the final image would act as a standalone memory cue. While viewing the scene description alongside the images during the internal evaluation may have made the association feel more explicit, this does not reflect the intended real-world use case. Ideally, the image would be stored anonymously on the user's device or printed and displayed discreetly, blending in with other personal media to avoid revealing its intended purpose.

Overall, the image-generation component showed promise, but it is important to acknowl-

edge key limitations:

This project did not include empirical testing with real users to measure memory performance across groups with and without visual memory aids. While the link between visual association and memory retention is strong, and the internal testing results are encouraging, the effectiveness of the image-based memory aid remains unverified. This represents a significant area for future research.

Additionally, the visual style of the generated images was purposely suited to a younger audience. Many scenes included animals and fantasy-inspired imagery, which aligned well with the tool's target age during development. However, if the application were to be adapted for more experienced or adult users, the visual design could shift toward more abstract or thematically mature representations. An age-specific filter or configurable style setting, allowing for minor scene peronsalisation, could be considered for future versions.

With any AI system that creates content with limited human oversight, responsible deployment requires active safeguards. This includes: continued prompt refinement, content auditing, real-time filtering where feasible, and user reporting mechanisms. While no inappropriate images were observed during initial internal testing, the unpredictable nature of generative models means the risk cannot be overlooked. Future versions of PassPOWER should include strategies for continuously evaluating the performance and safety of the generative models, especially if deployed in environments where content cannot be manually reviewed before display.

4.2.5 Educational Integration and Literacy Gaps

Although not evaluated during this phase of testing, the education component researched for the project remains central PassPOWER's broader purpose. This module can directly address identified gaps in password education by introducing users to core security principles through interactive content. By embedding guidance within the password creation process, it has the potential to transform passive tool usage into an active learning experience that builds lasting security habits.

4.2.6 Future Directions for AI and Tool Enhancement

During the development of this tool, OpenAI announced **gpt-image-1**, a new API model which allows creation of images across diverse styles, faithfully following custom guidelines, and leveraging world knowledge (OpenAI, 2024). As generative models continue to advance in quality and controllability, future versions of this application may benefit from tighter integration with such systems, offering more relevant, customizable, and secure visual representations. Ongoing awareness of these developments will be essential for maintaining the tools' effectiveness and long-term relevance.

CHAPTER 5

Conclusion

5.1 Introduction

This report presents the design and development of PassPOWER, an AI-assisted educational tool aimed at improving secure password practices for younger and less experienced users. By combining AI-assisted passphrase generation, strength evaluation, narrative scene generation, and prompt-based image rendering, the tool's development explored innovative methods to improve password strength and complex password memorisation through design choices aligned with modern security and learning principles. Although user testing and educational features remain areas for future work, the project demonstrates a working example of a more engaging and secure approach to teaching secure passphrase creation and retention.

5.2 Summary of Research

The research followed a structured, layered process to address the challenges inexperienced users face in developing and retaining strong passwords. These behaviours were examined alongside current NIST recommendations, prioritising password length and unpredictability over outdated complexity rules.

To improve memorability, the project explored insights from cognitive psychology such as *Dual Coding Theory*, which supports using verbal and visual cues in memory retention. This informed the inclusion of generated narrative scenes to embed passphrase content and the evaluation of modern image generation tools for producing age-appropriate visual cues.

In parallel, the study considered alternative education methods such as gamification and visual learning strategies to increase user engagement and reinforce secure password

concepts.

Ethical concerns concerning the use of generative AI were also examined, including the risks of generating inappropriate content, model transparency and long-term user reliance.

Together, these elements informed the development of PassPOWER, a tool designed to support secure password creation through the combination of educational design, memory reinforcement and responsible AI use.

5.3 Research Objectives

This project successfully addressed the following objectives:

1. **Identify common misconceptions about password security in young (13+) and inexperienced users.**

   Achieved through an in-depth literature review that examined user behaviour, educational gaps, and common password creation mistakes among youths and novice users.

2. **Develop a secure password generation and evaluation module that meets current industry complexity recommendations.**

   Implemented a passphrase generator aligned with NIST recommendations, using boh curated wordlist and AI-generative methods, with strength validated using *zxcvbn* and *Have I Been Pwnd?* databases.

3. **Integrate AI image generation and associated storytelling prompts, to deliver visual and cognitive reinforcement for password memorisation.**

   Built a system that transforms generated passphrases into descriptive scenes and corresponding AI-generated images to act as visual memory aids..

4. **Evaluate the effectiveness of AI-generated scenes and images in supporting memorisation of secure passphrases for age-appropriate users.**

   Conducted internal proof-of-concept testing to verify whether passphrase components were preserved in scene description and visuals, supporting their potential as memory cues.

5. **Identify and address potential ethical and security risks associated with password and image-generative AI tools.**

   Explored risks including inappropriate outputs, data privacy concerns, over-reliance on AI, and compliance with GDPR/COPPA through system design decisions and literature-informed discussion.

5.4 Research Contribution

While the individual components used in PassPOWER are well established, their integration into a unified, age-appropriate tool for secure passphrase generation and visual cue-based memorisation represents a practical contribution to applied cybersecurity education and design.

5.5 Future Work

Building on the findings discussed in Chapter 4, several clear directions for future development and evaluation of PassPOWER have been identified.

First, a key limitation of this project was the absence of live user testing. Future work should include empirical studies with real users, especially across younger age groups, to evaluate the effectiveness of AI-generated imagery as a memory aid. These studies should aim to determine whether the system improves recall and the appropriate minimum age for independent use.

Secondly, while the educational component was designed, it was not included in the internal testing phase. Expanding and evaluating this module represents a high-priority opportunity. Once implemented, it could elevate PassPOWER from a proof-of-concept into a comprehensive education platform that promotes password security comprehension and secure online habits.

Third, the reliance on AI-assistance for passphrase generation introduced some uncertainty around entropy and true randomisation of passphrase word components. Future versions should explore the integration of a larger, audited wordlist to support transparent entropy calculations and ensure compliance with NIST's 64-bit entropy recommendations. Comparative testing between AI-assisted and static wordlist-based generation could help determine the best balance between memorability, transparency and overall security.

Fourth, as highlighted during image testing, prompt behaviour and model interpretability remain areas of risk. Future versions should include mechanisms for ongoing content auditing, real-time prompt evaluation, and user reporting to ensure all outputs remain age-appropriate and effective. Integration of newer models such as gpt-image-1 may offer improved results between prompt and output, as well as increased stylistic control.

In addition, future development should explore ways to minimise or eliminate the need to send sensitive data, such as passphrases or descriptive prompts, to external APIs. Hosting PassPOWER on a private server with strict data handling policies and exploring options for local or on-device AI models could mitigate privacy concerns, especially in educational contexts involving children.

Finally, adapting the visual style and scene narratives to suit older or more experienced user groups could expand PassPOWER's application possibilities. While the potential for

creative and personalised outputs is significant, future development must be balanced by safeguards that ensure content remains age-appropriate, secure, and trustworthy.

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APPENDIX A

Survey Consent Form Body

Consent to Participate in Password Memory Study

You are being invited to take part in a short memory study. Please review the information below before deciding whether to participate.

What You Will Do:

- You will be given a password to memorize.
- Depending on your assigned group, you may also receive a short scene description and/or an image.
- You will be contacted within the next few days and asked to recall the password from memory.
- If provided with an image, you are allowed to save and refer to it during the recall attempt. Otherwise, you must rely on memory alone.

Risks and Benefits:

- Risks are minimal and no personal information will be collected.
- Participation may help researchers understand memory and password strategies.

Your Rights:

- Participation is voluntary.
- You may withdraw at any time without penalty.
- Your responses will remain confidential and used only for study purposes.

Consent: By participating, you acknowledge that:

- You have read and understood the information above.
- You voluntarily agree to participate.
- You understand that you may be contacted for password recall within the next few days.

Name: _____

Date: _____

Figure A.1: Survey Consent Form presented to participants.

APPENDIX B

Survey Group A Email Body

Your password: `helmet-tiger-garden`

Take a moment to memorize your password.

To support your memory, here is a simple scene description:

Wearing a helmet, Jake cautiously entered the garden where the circus tiger had been spotted.

Here is a visual memory cue based on the provided scene:



Important:
The image has also been provided as an attachment to this email, please save it to your device.

When you are contacted in the coming days to recall the password, this image will be the only aid you are allowed to refer to. You may use it to help trigger your memory.

Once you have memorized the password and created a mental connection with the scene and image, **please dispose of the password and scene description.** Retain only the image.

Figure B.1: Instructions email sent to Group A participants.

APPENDIX C

Survey Group B Email Body

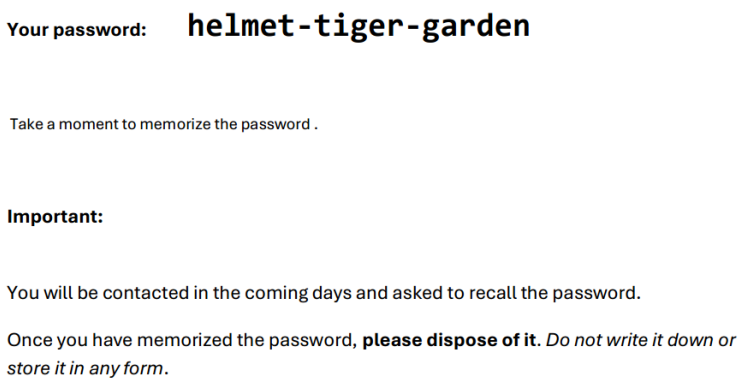


Figure C.1: Instructions email sent to Group B participants.

APPENDIX D

PassPOWER Test Table

#	Generated Passphrase	Character Count (Not including hyphen)	Generated Scene Description	zxcvbn Score (0–4)	Generated Image	Notes
1	backpack-cookie-giraffe	21	In a sunlit savanna, a curious giraffe peeks into a teenagers backpack, gently snoathcing a cookkie with its long tongue, much to the delight of the surrounding giggling children.	4		Clear image. Giraffe and cookie well depicted. Backpack visible but detail could be sharper. Strong scene match.
2	balloon -apple-monster	19	A red balloon floated above the trees while a moster playfully juggled apples nearby.	4		Fun, vibrant scene. Monster and apples clear. Balloon present. Minor artistic abstraction.
3	calculator-hedgehog-mountain	26	In a vibrant meadow at the foot of a towering mountain, a small hedgehog, calculator in hand, enthusiacitally tallies the flowers he needs to plat to keep the environment flourishing.	4		Hedgehog and calculator clearly shown. Mountain present. Vibrant composition.
4	guitar-Kraken-Brocolli	20	At the beach campfire, Jamie stremmed his guiar while everyone joked about a kraken hiding among the broccoli in their dinner salads.	4		Guitar clear. Kraken and broccoli represented, though kraken could be misinterpreted. Kraken noted as a potential unknown object.
5	hamburger-dolphin-notebook	24	At the sunlit pier, Max scribbled in his notebook about the dolphin he'd just fed a hamburger, both of them sharing a magical moment of friendship.	4		Dolphin and hamburger clearly rendered. Notebook was small but visible.
6	helmet-tiger-garden	17	Wearing a helmet, Jake cautiously entered the garden where the circus tiger had been spotted.	4		Helmet and tiger very clear. Garden was not as clear, needs user interpretation.
7	penguin-flower-elephant	21	A penguin waddled past an elephant sniffing a colorful flower in the zoo.	4		Penguin and elephant and flower were all clear. Good clarity and word-to-image representation.
8	Saxophone-Goblin-Triceratops	26	At the funky forest party, a goblin skillfully plaed the saxophone while riding on a friendly triceratops.	4		Wild scene with good clarity. Goblin and triceratops and saxophone distinctly represented.
9	moth-subway-ticket	16	A moth fluttered around the dim light as the teenager held his subway ticket tightly, waiting for the train.	4		Confusion on subway car. Abstract representation, garden growing inside subway car. Ticket and moth represented clearly.
10	sunset-waffle-monkey	18	As the sunset painted the sky orange, a mischievous monkey snatched a waffle right out of a teenager's hand.	4		Monkey and waffle very clear. Sunset background vibrant. All elements present and well rendered.

Figure D.1: Evaluation of PassPOWER Passphrase-to-Scene/Image Generation (10 Examples). Evaluation conducted April 30, 2025 using OpenAI ChatGPT-4-Turbo for passphrase and scene description, and DALL·E 3 for image generation.

APPENDIX E

Generated Passphrase Images

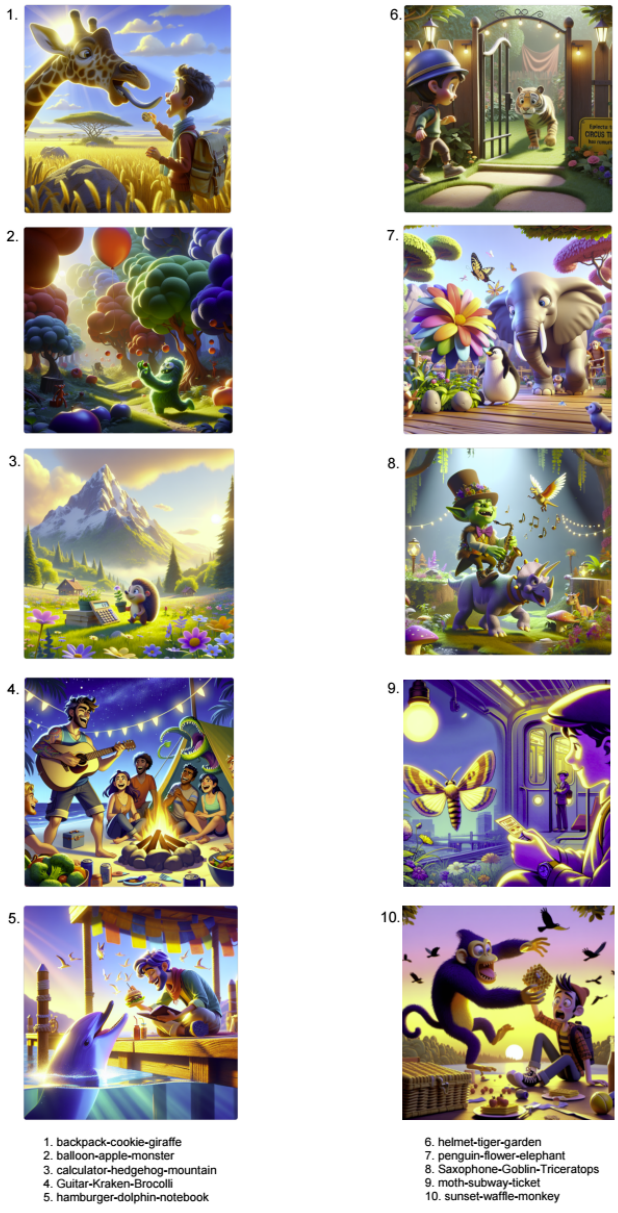


Figure E.1: Hi-Res Generated Passphrase Images from PassPOWER Test Table

APPENDIX F

PassPOWER GitHub Repo

The source code for the PassPOWER project can be found at the following GitHub repository. See README.md for instructions:

https://github.com/evylk/studio2/tree/PassPower_Chat_Version

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 10775 Words in text: 10237 Words in headers: 221 Words outside text (captions, etc.): 314 Number of headers: 60 Number of floats/tables/figures: 19 Number of math inlines: 0 Number of math displayed: 3
(errors:1) NOTE: References are excluded.